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Supplemental Material

The Shape of the Concentration–Response Association between Fine Particulate Matter Pollution and Human Mortality in Beijing, China, and Its Implications for Health Impact Assessment

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Figure S6. Percentage change (95% confidence intervals) in daily mortality per 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ at lag01, at different levels of $\text{PM}_{2.5}$, using observed data of the monitors in Hai Dian Bao Lian (HDBL) station and U.S. Embassy. The shape of the point estimates refers to different models, with estimates from fitting the linear model shown with circles, the piecewise-linear model with triangles, and the spline model with squares. Results were estimated using generalized linear models with over-dispersed Poisson distributions, controlling for natural cubic splines for time trends (7df/year), daily mean temperature (3df) and daily relative humidity (3df), and indicator terms for day of the week and holidays.

Figure S7. Concentration-response curves for non-accidental (top), circulatory (middle), and respiratory mortality (bottom) fit using a linear (left), piecewise-linear (middle), and spline model (right) for lag01 days for $\text{PM}_{2.5}$ using year-round data. Dotted lines show 95% confidence intervals for each model fit. Tick marks on the bottoms of the graphs show the distribution of $\text{PM}_{2.5}$ concentrations across study days. RRs were estimated using generalized linear models with over-dispersed Poisson distributions, controlling for natural cubic splines for time trends (7 d.f./year), daily mean temperature (3 d.f.) and **daily dewpoint temperature (3 d.f.)**, and indicator terms for day of the week and holidays.

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	Year 2009	Year 2011
Sampled total population (person)	15,096	17,246
<i>By gender</i>		
Male	7,705 (51%)	8,796 (51%)
Female	7,390 (49%)	8,450 (49%)
<i>By age (years)</i>		
Aged 0–14	1,497 (10%)	1,510 (9%)
Aged 15–64	12,075 (80%)	14,216 (82%)
Aged ≥65	1,524 (10%)	1,521 (9%)
Sampled total number of households (household)	5,444	6,166
1 person per household	1,013 (19%)	1,541 (25%)
2 – 3 persons per household	3,597 (66%)	3,688 (60%)
≥ 4 persons per household	833 (15%)	937 (15%)

Table S2 Quasi-Akaike's Information Criterion (Q-AIC) for linear, piecewise-linear, and spline models. Results are based on generalized linear models with over-dispersed Poisson distributions, controlling for natural cubic splines for time trends (7 d.f./year), daily mean temperature (3 d.f.) and daily relative humidity (3 d.f.), and indicator terms for day of the week and holidays.

	Non-accidental mortality	Circulatory mortality	Respiratory mortality
Linear model	12387.12	11280.77	8530.16
Piecewise-linear model	12389.74	11292.59	8522.01
Spline model	12384.18	11272.03	8528.44

Table S3 Estimated number and fraction of deaths attributable to PM_{2.5} in Beijing on average for the six months of the cold or warm season within a year (based on 2009–2012) compared to counterfactual scenarios in which no study day exceeds either (1) a background PM_{2.5} concentrations of 7.5 µg/m³, (2) the WHO daily guideline of 25 µg/m³, or (3) the CAAQS 24-hour standard of 75 µg/m³. Estimated excess deaths are calculated using concentration-response functions generated using linear, piecewise-linear, or spline models fit to year-round data with over-dispersed Poisson distributions, controlling for natural cubic splines for time trends (7 d.f./year), daily mean temperature (3 d.f.) and daily relative humidity (3 d.f.), and indicator terms for day of the week and holidays. Numbers in parentheses show 95% confidence intervals for either attributable number or attributable fraction.

		Counterfactual PM _{2.5} threshold					
		7.5 µg/m ³		25 µg/m ³		75 µg/m ³	
		Attributable number	Attributable fraction (%)	Attributable number	Attributable fraction (%)	Attributable number	Attributable fraction (%)
Non-accidental mortality							
<i>Cold-season</i>							
	Linear model	593 (209, 959)	1.55 (0.53, 2.53)	469 (181, 767)	1.22 (0.45, 1.99)	214 (75, 351)	0.56 (0.21, 0.92)
	Piecewise-linear model	263 (-278, 800)	0.54 (-0.99, 2.00)	246 (-143, 642)	0.55 (-0.52, 1.63)	209 (74, 346)	0.57 (0.21, 0.92)
	Spline model	196 (-569, 938)	0.51 (-1.45, 2.43)	210 (-266, 679)	0.55 (-0.72, 1.76)	204 (44, 363)	0.53 (0.12, 0.94)
<i>Warm-season</i>							
	Linear model	441 (162, 732)	1.37 (0.50, 2.21)	336 (114, 547)	1.05 (0.37, 1.72)	108 (39, 178)	0.34 (0.12, 0.55)
	Piecewise-linear model	116 (-370, 590)	0.20 (-1.37, 1.74)	101 (-247, 450)	0.21 (-0.92, 1.32)	69 (-15, 154)	0.23 (-0.01, 0.47)
	Spline model	63 (-605, 710)	0.19 (-1.81, 2.25)	74 (-340, 499)	0.23 (-1.13, 1.54)	68 (-30, 163)	0.21 (-0.08, 0.51)
Circulatory mortality							
<i>Cold-season</i>							
	Linear model	471 (203, 739)	2.37 (1.00, 3.72)	374 (159, 586)	1.88 (0.83, 2.93)	171 (75, 269)	0.86 (0.36, 1.34)
	Piecewise-linear model	121 (-278, 506)	0.36 (-1.69, 2.39)	135 (-160, 417)	0.53 (-1.00, 2.01)	164 (67, 259)	0.88 (0.40, 1.36)
	Spline model	16 (-537, 562)	0.08 (-2.72, 2.78)	80 (-275, 417)	0.40 (-1.35, 2.10)	166 (53, 277)	0.84 (0.25, 1.37)
<i>Warm-season</i>							
	Linear model	323 (137, 504)	2.09 (0.91, 3.28)	247 (106, 388)	1.60 (0.68, 2.54)	80 (34, 125)	0.52 (0.23, 0.80)
	Piecewise-linear model	6 (-320, 333)	-0.23 (-2.44, 1.90)	17 (-266, 255)	-0.06 (-1.66, 1.46)	41 (-16, 96)	0.31 (-0.03, 0.63)
	Spline model	-75 (-519, 356)	-0.49 (-3.40, 2.33)	-25 (-312, 262)	-0.16 (-2.00, 1.68)	46 (018, 107)	0.30 (-0.11, 0.70)
Respiratory mortality							
<i>Cold-season</i>							
	Linear model	65 (-50, 167)	1.61 (-1.11, 4.14)	51 (-35, 131)	1.27 (-0.85, 3.31)	23 (-16, 60)	0.57 (-0.39, 1.51)
	Piecewise-linear model	132 (-26, 281)	3.48 (-0.53, 7.36)	97 (-16, 203)	2.53 (-0.30, 5.27)	24 (-14, 62)	0.56 (-0.41, 1.49)
	Spline model	148 (-68, 354)	3.67 (-1.76, 8.65)	109 (-26, 240)	2.71 (-0.65, 5.85)	30 (-16, 73)	0.75 (-0.41, 1.83)
<i>Warm-season</i>							
	Linear model	46 (-30, 121)	1.45 (-0.96, 3.80)	35 (-24, 92)	1.10 (-0.73, 2.86)	11 (-8, 29)	0.35 (-0.22, 0.93)
	Piecewise-linear model	108 (-24, 226)	3.62 (-0.63, 7.61)	80 (-12, 175)	2.66 (-0.38, 5.59)	19 (-3, 41)	0.55 (-0.14, 1.22)
	Spline model	121 (-55, 285)	3.82 (-1.73, 9.16)	90 (-23, 199)	2.85 (-0.70, 6.35)	24 (-2, 48)	0.75 (-0.06, 1.56)

Note: Since some models estimated relative risks lower than 1 within some concentration ranges of PM_{2.5}, some scenarios resulted in an estimated deficit of deaths (negative values in table cells) rather than estimated excess deaths compared to the counterfactual.

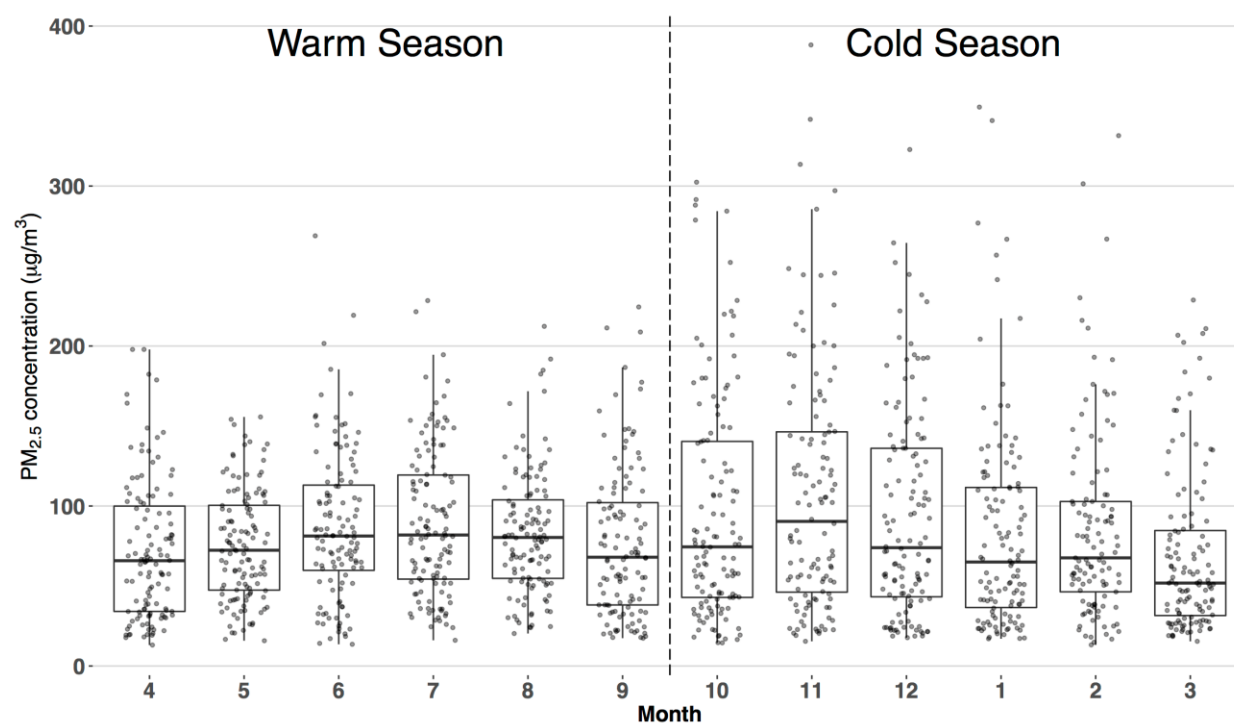


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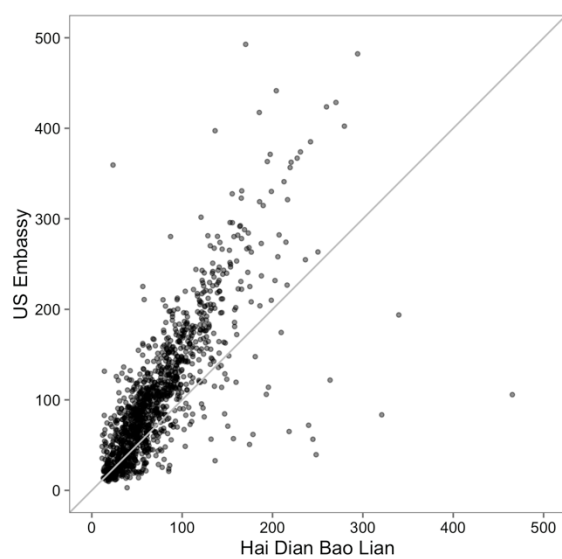


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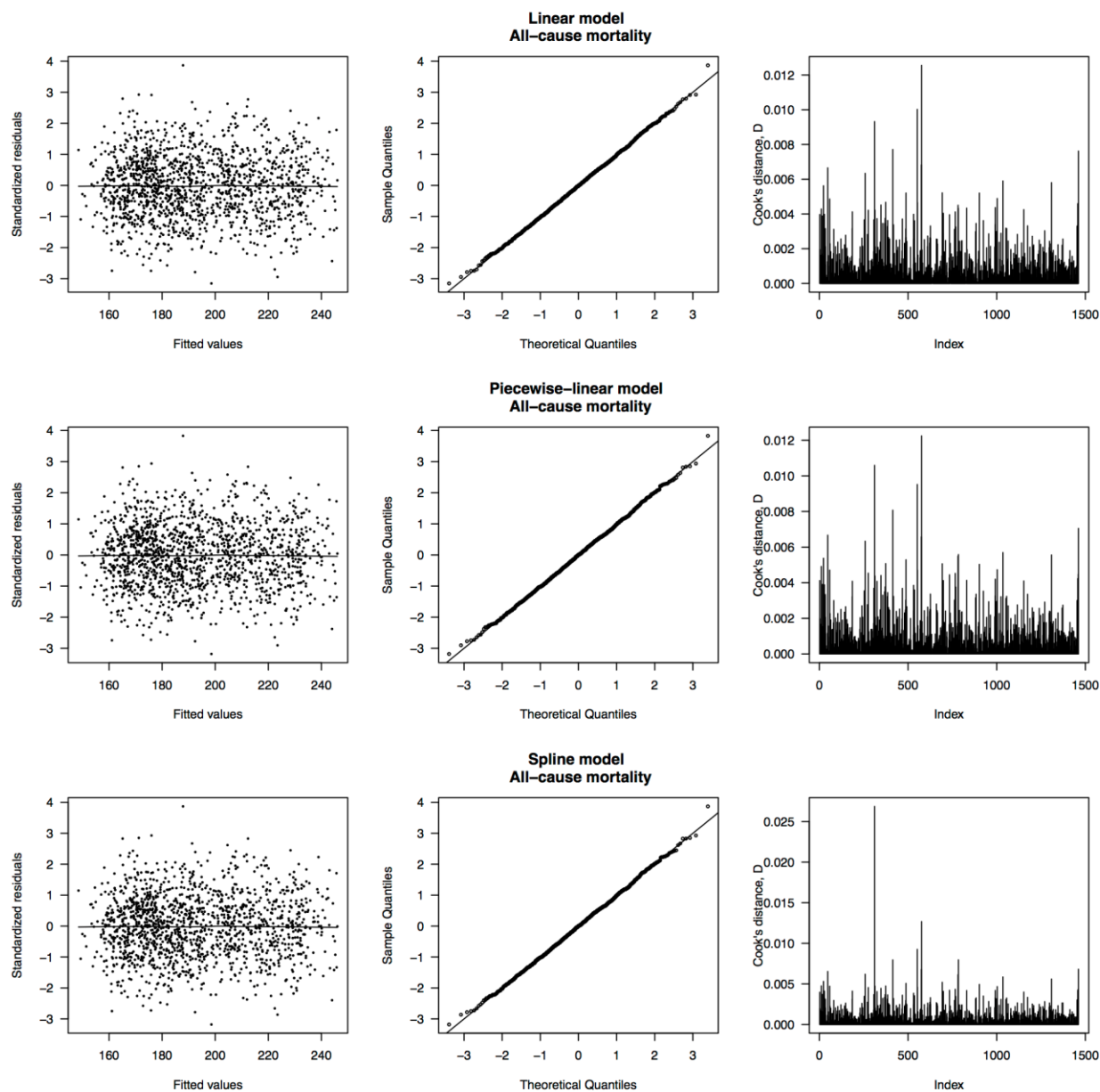


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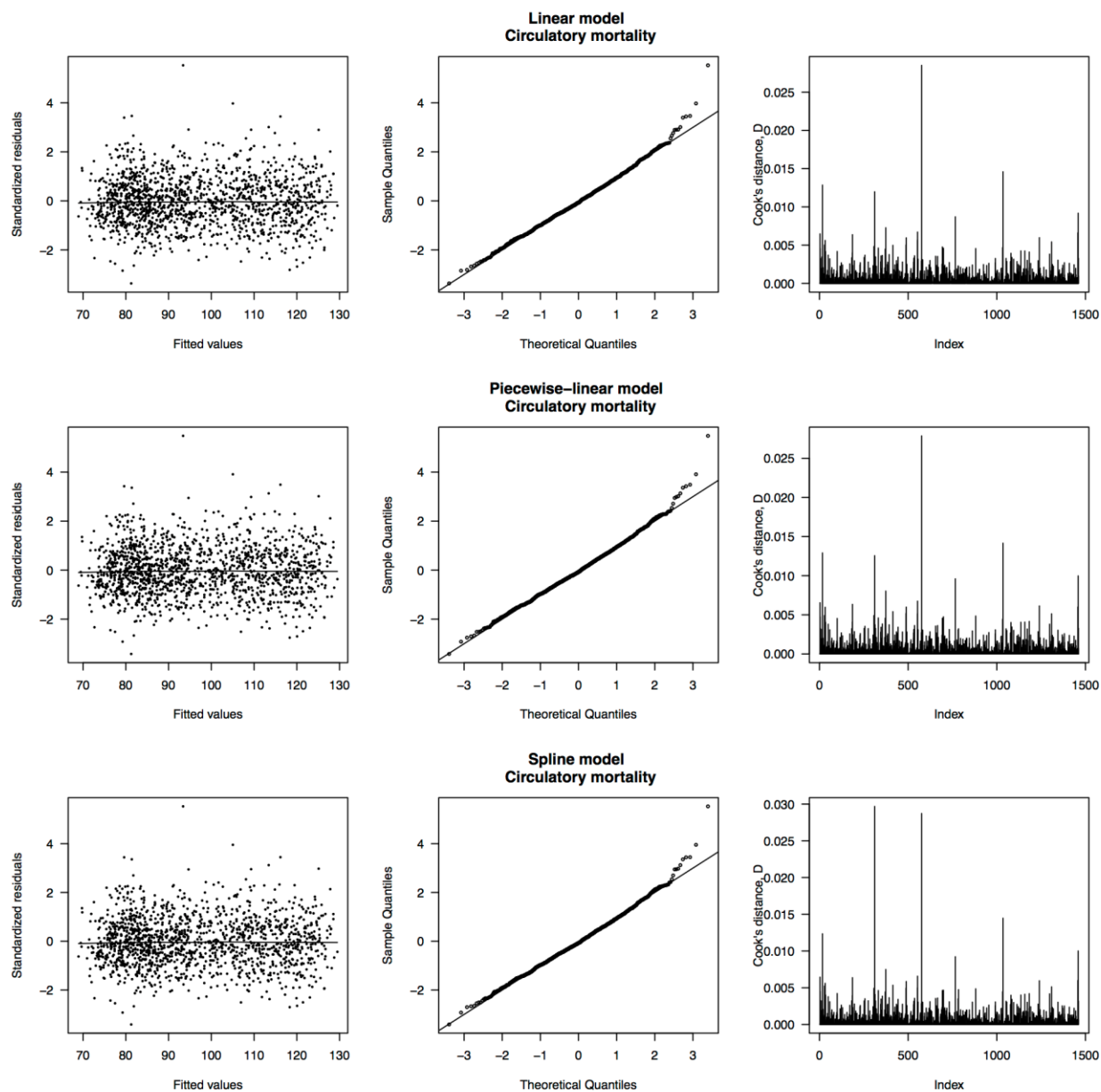


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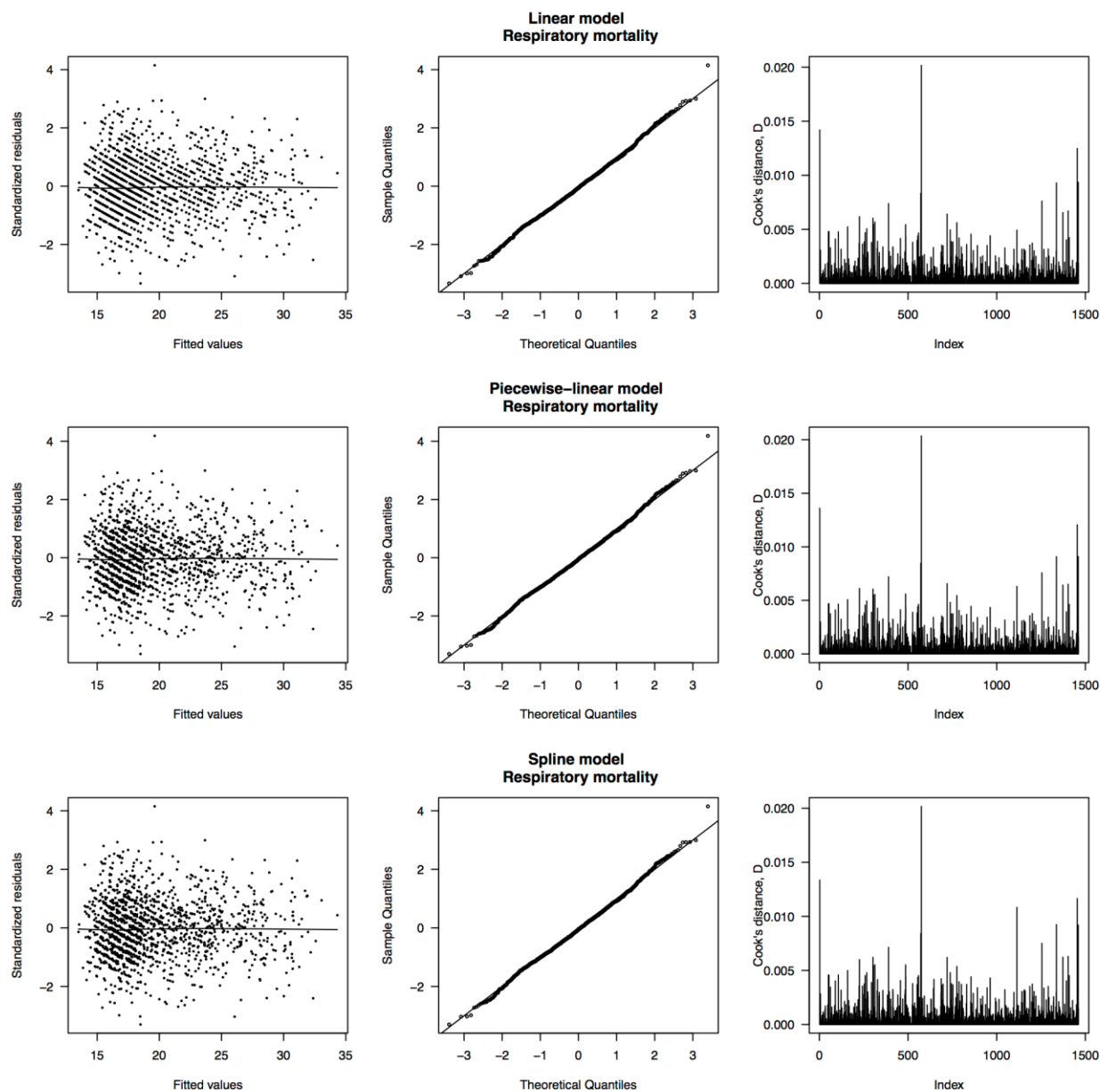


Figure S5 Diagnostic plots for modeling C-R associations between $PM_{2.5}$ and respiratory mortality using linear (top), piecewise-linear (middle), and spline (bottom) models. This figure shows standardized residuals (left), Q-Q plots of standardized residuals (middle), and Cook's distance measurements (right). Results are based on generalized linear models with over-dispersed Poisson distributions, controlling for natural cubic splines for time trends (7 d.f./year), daily mean temperature (3 d.f.) and daily relative humidity (3 d.f.), and indicator terms for day of the week and holidays. There are 3, 4, and 4 outlier observations (standardized residual greater than 3) for linear, piecewise-linear, and spline model, respectively.

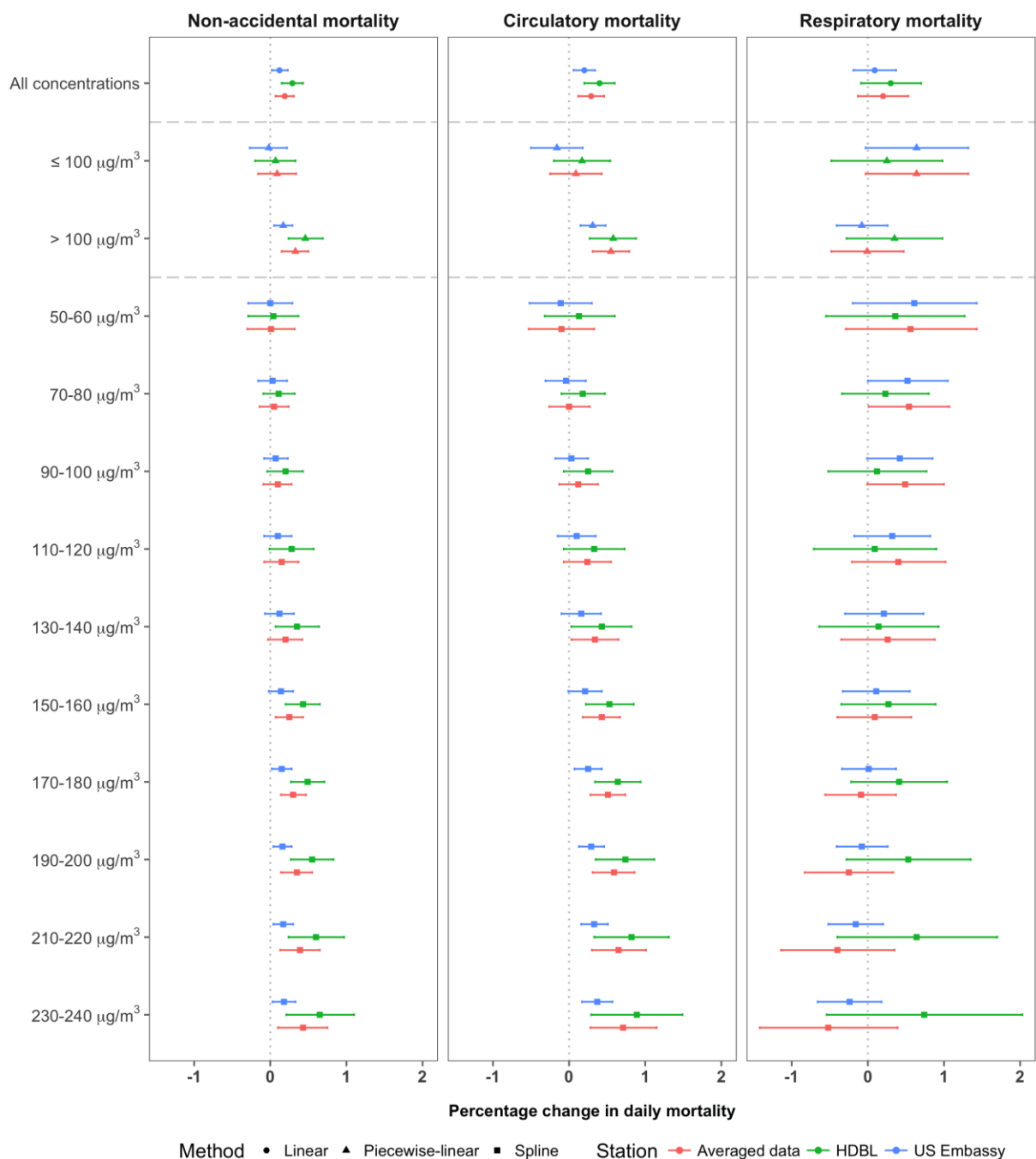


Figure S6 Percentage change (95% confidence intervals) in daily mortality per 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ at lag01, at different levels of $\text{PM}_{2.5}$, using observed data of the monitors in Hai Dian Bao Lian (HDBL) station and U.S. Embassy. The shape of the point estimates refers to different models, with estimates from fitting the linear model shown with circles, the piecewise-linear model with triangles, and the spline model with squares. Results were estimated using generalized linear models with over-dispersed Poisson distributions, controlling for natural cubic splines for time trends (7df/year), daily mean temperature (3df) and daily relative humidity (3df), and indicator terms for day of the week and holidays.

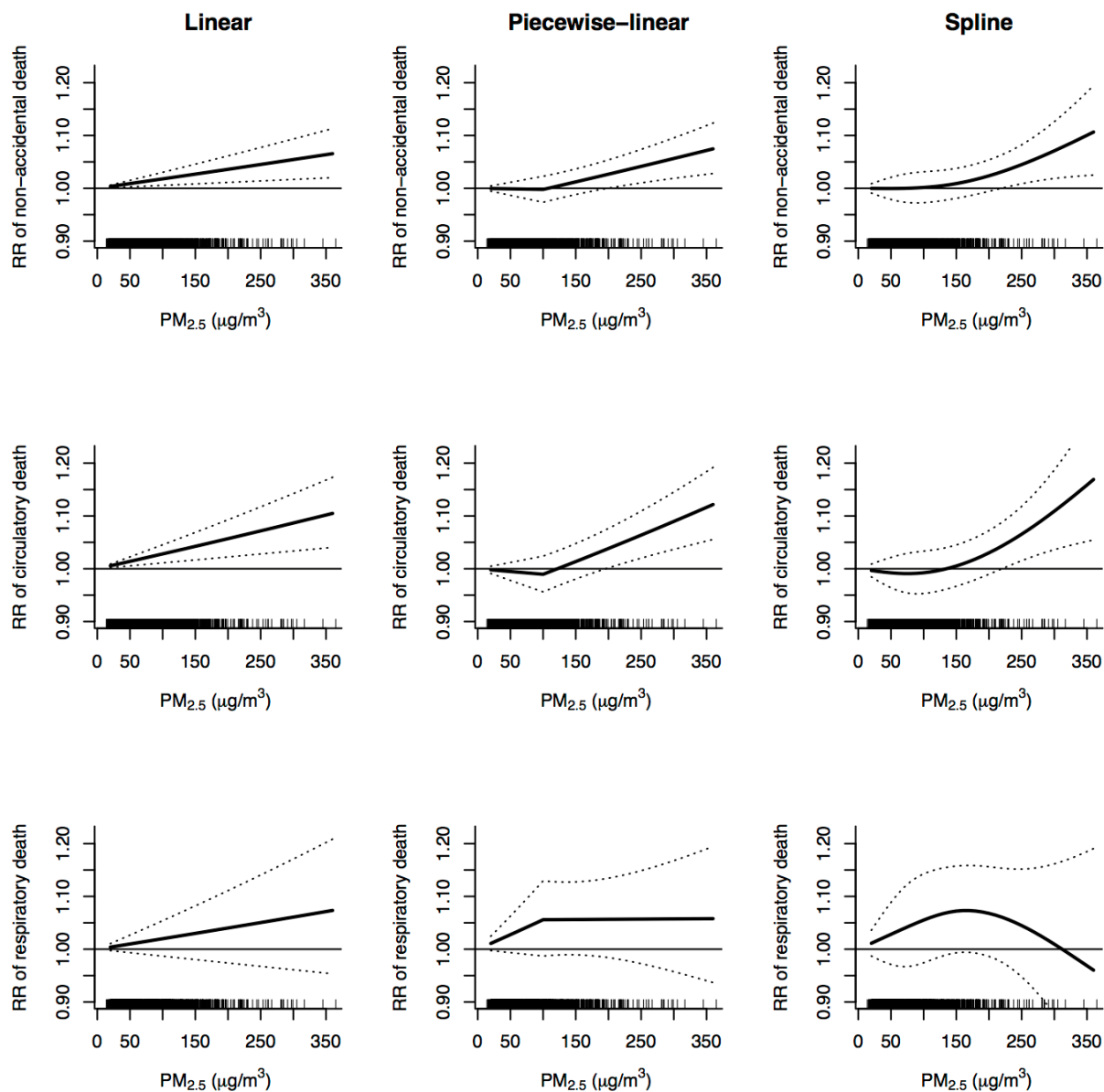


Figure S7 Concentration-response curves for non-accidental (top), circulatory (middle), and respiratory mortality (bottom) fit using a linear (left), piecewise-linear (middle), and spline model (right) for lag01 days for $PM_{2.5}$ using year-round data. Dotted lines show 95% confidence intervals for each model fit. Tick marks on the bottoms of the graphs show the distribution of $PM_{2.5}$ concentrations across study days. RRs were estimated using generalized linear models with over-dispersed Poisson distributions, controlling for natural cubic splines for time trends (7 d.f./year), daily mean temperature (3 d.f.) and **daily dewpoint temperature (3 d.f.)**, and indicator terms for day of the week and holidays.

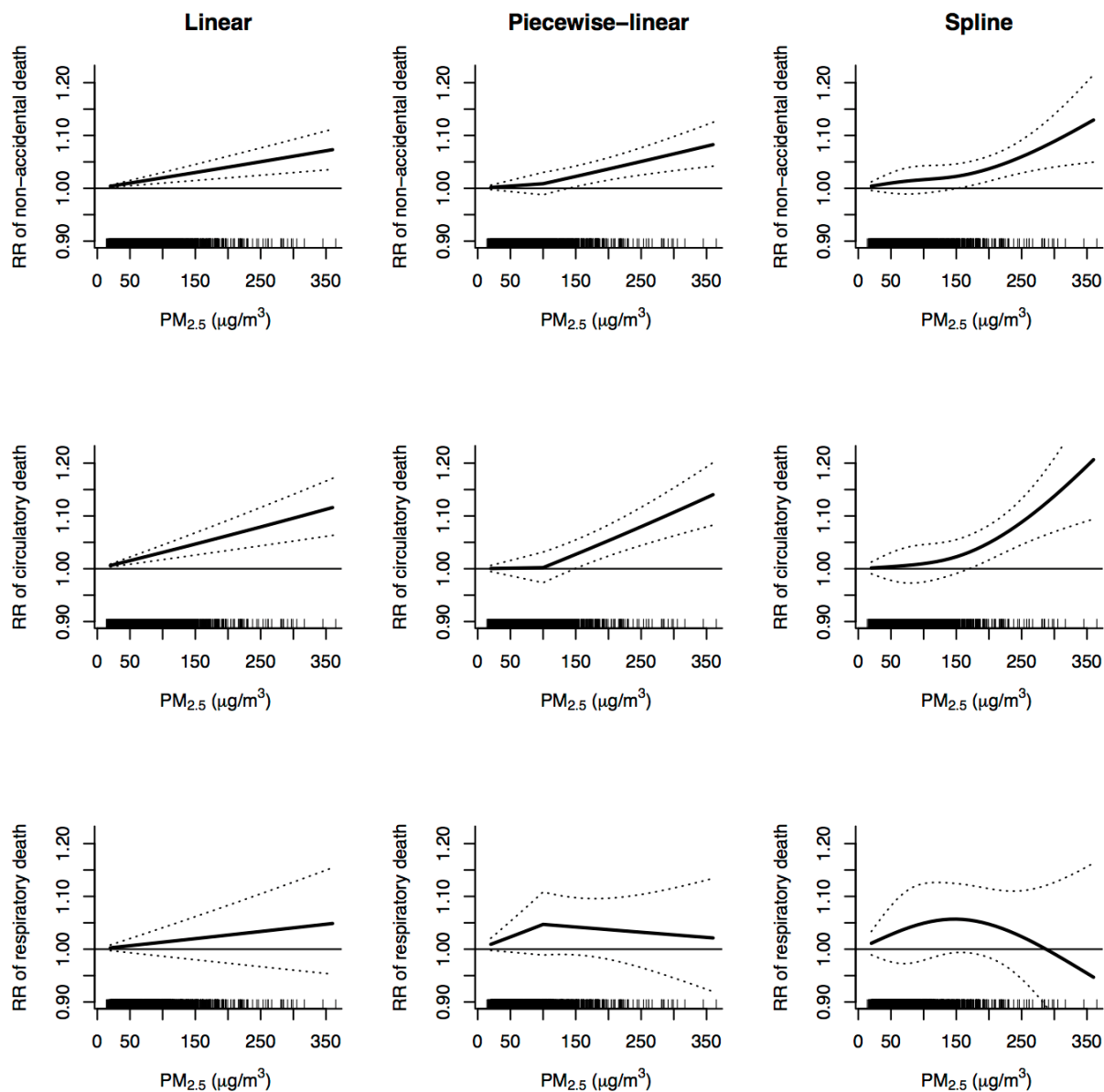


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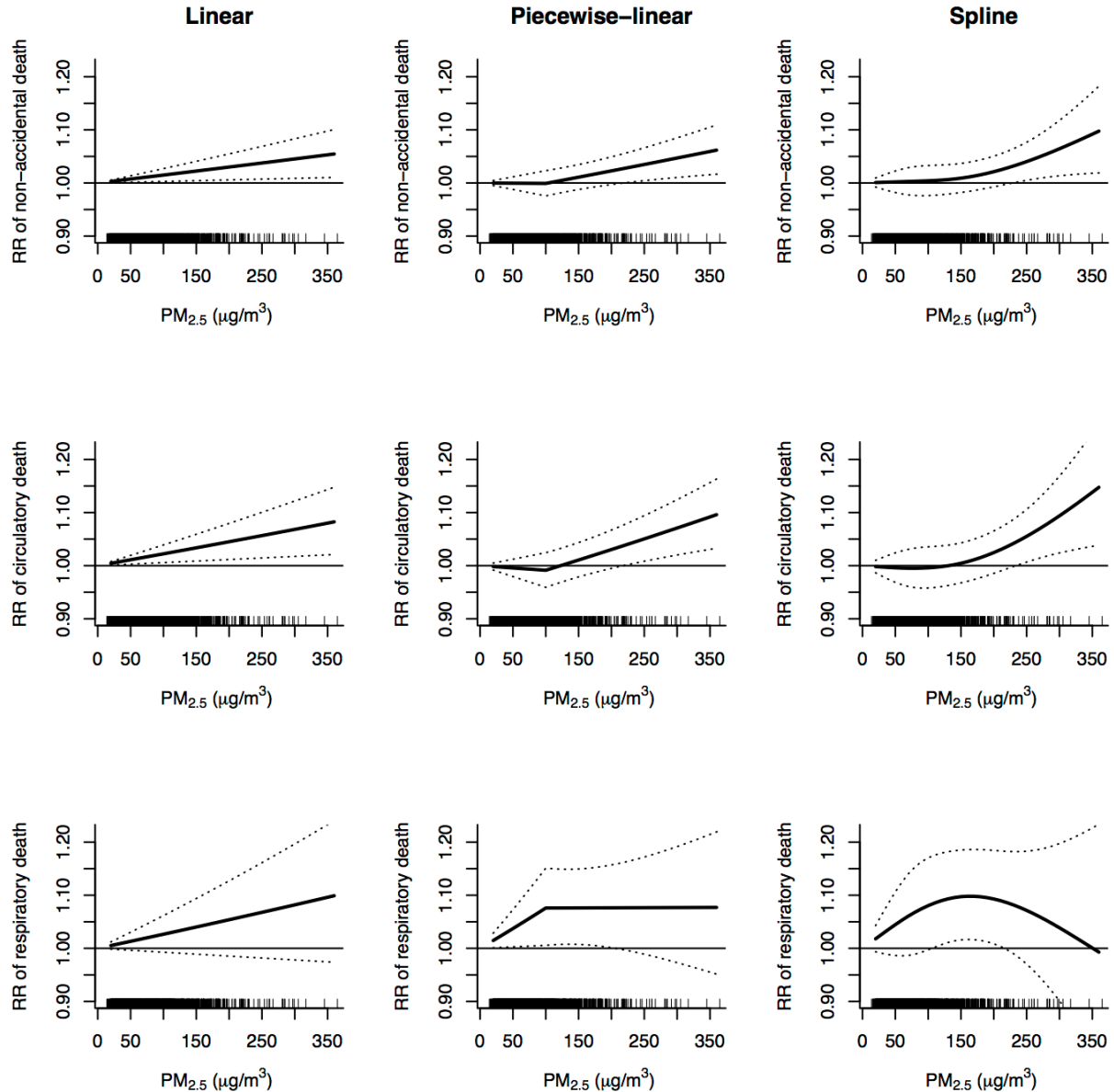


Figure S9 Concentration-response curves for non-accidental (top), circulatory (middle) and respiratory mortality (bottom) fit using a linear (left), piecewise-linear (middle), and spline model (right) for $PM_{2.5}$ at lag01 using year-round data. Dotted lines show 95% confidence intervals for each model fit. Tick marks on the bottoms of the graphs show the distribution of $PM_{2.5}$ concentrations across study days. RRs were estimated using generalized linear model with over-dispersed Poisson distributions, controlling for natural cubic splines for time trends (7 d.f./year) and daily relative humidity (3 d.f.), **a constrained distributed lag function of daily mean temperature up to 21 preceding days**, and indicator terms for day of the week and holidays.

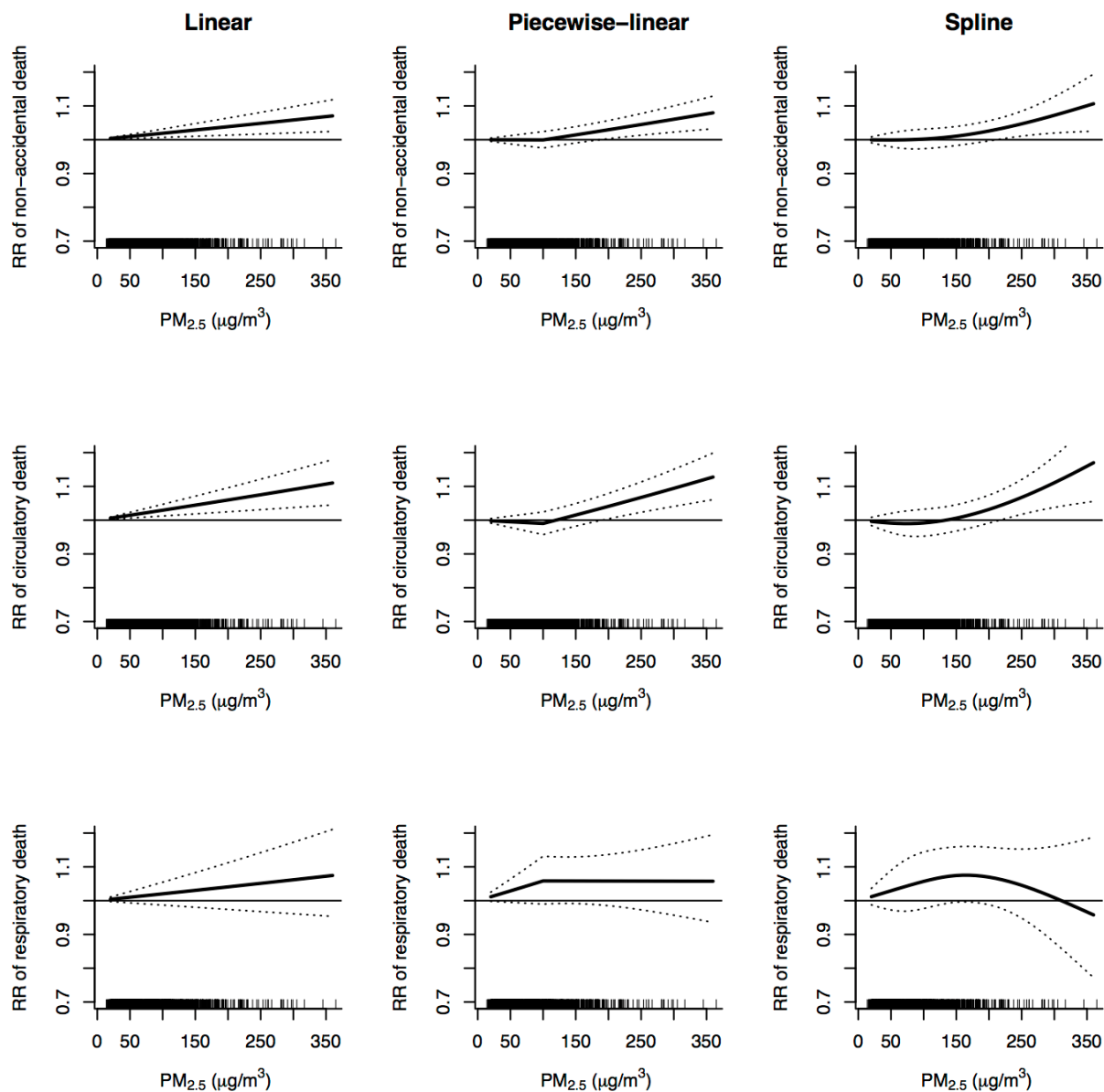


Figure S10 Concentration-response curves for circulatory (top) and respiratory mortality (bottom) fit using a linear (left), piecewise-linear (middle), and spline model (right) for **lag02 days for PM_{2.5}** using year-round data. Dotted lines show 95% confidence intervals for each model fit. Tick marks on the bottoms of the graphs show the distribution of PM_{2.5} concentrations across study days. RRs were estimated using generalized linear models with over-dispersed Poisson distributions, controlling for natural cubic splines for time trends (7 d.f./year), daily mean temperature (3 d.f.) and daily relative humidity (3 d.f.), and indicator terms for day of the week and holidays.